Title: Generative Modeling of Toronto Bird Songs Using VAEs and Diffusion Models

This project investigates whether deep generative models can learn to generate realistic bird songs for a small set of species commonly found in the Toronto area. We model bird vocalizations as samples from an unknown species-specific distribution and train models to sample from it. Instead of generating raw audio directly, each recording will be converted into a fixed-size log-mel spectrogram, allowing the audio-generation problem to be treated as an image-generation task. Generated spectrograms will then be converted back into audio using a vocoder.

We will compare two deep generative model families: a convolutional Variational Autoencoder and a denoising diffusion model. The VAE will learn an encoder-decoder representation with a variational latent space, using a reconstruction loss and KL-divergence regularization. New samples will be generated by sampling from the learned latent space. The diffusion model will learn a forward noising process and a reverse denoising process. We expect the VAE to train more efficiently but produce smoother or blurrier spectrograms, while the diffusion model may generate sharper and more realistic samples at the cost of longer training time.

The main dataset will be Bird Song Data Set found at vinayshanbhag/bird-song-data-set on Kaggle. We will focus on three Toronto-area species with clear songs, such as American Robin, Northern Cardinal, and Song Sparrow, which are commonly found in the Toronto area. The dataset is suitable for this project because it excludes non-song vocalizations such as alarm calls and scolding calls, includes only highest-quality Xeno-canto recordings, and provides standardized 3-second mono WAV clips sampled at 22,050 Hz. Preprocessing will focus on verifying file consistency, selecting the three target species, and converting each clip into a normalized 128×128 log-mel spectrogram. We will check for corrupted files, confirm that clips have the expected duration and sampling rate, and inspect class balance across the selected species.

The project outputs will include a curated bird-song dataset, a reproducible preprocessing pipeline, trained VAE and diffusion models, generated spectrograms, reconstructed audio samples, and a **comparative evaluation**. We will first validate the full pipeline on one species before scaling to the full three-species experiment. Training will be performed on a desktop RTX 3070 Ti with 8 GB of VRAM, using compact model architectures, mixed-precision training, and gradient accumulation if needed.

Evaluation will combine quantitative and qualitative methods. For the VAE, we will report held-out reconstruction errors and show real-versus-reconstructed spectrograms. For both models, we will compute an FAD-style distributional distance between generated and real audio embeddings. FAD is an audio-domain analogue of FID that compares generated and reference audio distributions in a learned embedding space. To make this metric more interpretable, we will include a real-vs-real baseline and a vocoder reconstruction baseline, where real audio is converted to a spectrogram and reconstructed back to audio. This will help separate generative-model errors from spectrogram inversion artifacts.

Species recognizability will be evaluated using either a small in-house classifier trained on real spectrograms or external tools like Merlin Bird ID and BirdNET, which supports automated bird-sound identification and can output species predictions and embeddings. To assess diversity and memorization, we will compare generated-to-training nearest-neighbor distances with held-out-test-to-training distances, and compute pairwise embedding-distance distributions for real and generated samples. This evaluation assesses whether the models generate diverse samples rather than memorizing the training data. Finally, a small blind listening test will compare real, vocoder-reconstructed, VAE-generated, and diffusion-generated clips based on realism, audio quality, and species identity.

This project connects to data distributions, sampling, VAEs, diffusion models, and conditional generation. As a stretch goal, we may train species-conditioned models, allowing the user to specify a species label and generate corresponding bird-song samples.